

# Course milestone M5: Pathway Declaration and Mandated Contrast

## HONR 46400 · Evidence-Driven Research

### Start here

Everything this milestone collects goes into one file, [lastname\\_m05.pdf](#). Build it in this order:

1. **Your carry-forward record, at the very top.** This is where you answer me. It holds two things you already have: every numbered request from my review of your last milestone (in my email and your Brightspace feedback, each with an ID like **IR-M05-01**), and every step your research action plan sets for this milestone (use the latest version I sent you). Next to each one, write **Done** and where I can find it, or **Not adopted**, with your reason and what you did instead.
2. **Your milestone work**, as the book milestone below describes it.

When a request, a plan step and the book work ask for the same thing, do it once and point to it from your record.

**How it is scored.** The course rubric on Brightspace scores this milestone. It already includes the book's own checks, plus two things for the course: your action-plan steps are one row of the rubric, and each numbered request left unanswered costs 5 points, up to 20. If I sent you no numbered requests, say so in one line and move on. Not sure whether something applies to you? Ask me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M05**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 5: Your pathway, declared** (checkpoint, version 1) of the course book, EDR|AI.

### Open the milestone workbook in Colab

**This chapter closes Studio 5: Develop the pathway.** Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

**What this milestone produces.** Research Contract v1: objective by target and reach by data strategy by warrant, with the pathway declared and its limits written before any result exists.

### What you bring

You also carry forward the last milestone's artifact: **Milestone 4: Your research contract, v0**. This one builds on it, and the chain assumes it is versioned and filed.

These pieces are conditional. Bring each one when its condition is yours; when it is not, the lesson's own drill or a plain skip is the right call, and the milestone does not require the piece.

- Lesson 14: Observational Descriptive Research: either your four boxes and selection diagram, if this is your pathway, or the classification drill that rules it out.
- Lesson 15: Observational Causal Research: either your causal diagram with its leverage named and priced, or the drill that shows why this road is not yours.
- Lesson 16: Experimental Descriptive Research: either your revealed-characteristic estimand with its demand and instrument threats named, or the drill run on your own question.
- Lesson 17: Prediction and Generalization: either your signed four-part forecasting contract with its boundary, or the contrast drill: the plain rejection line, the prediction-time leakage check run on your own features, and two sentences on why prediction answers a different question.
- Lesson 18: Experimental Causal Research: either your treatment, outcome, and potential outcomes with the nearest threat named, or the drill plus the confounder that blocks the causal reading.
- Lesson 19: Hybrid and Complex Designs: if your design has stages: the stage-by-stage map, its weakest link, and the one-sentence justification for keeping or cutting it.

One bound on the skip rule here: this milestone requires your PRIMARY pathway's full piece plus ONE contrast drill from any other route you choose: reading a road that is not yours is what makes yours visible; skipping the rest is correct. The overlay binds only when your design has stages.

## The practice

1. Use the route guide on the studio opener to choose your pathway, then read that lesson and at least one that does not fit, so you can say why yours does.
2. Declare your pathway and write the crossing that licenses it: what makes your comparison mean what you claim.
3. Write the limits: what this pathway cannot establish no matter how clean the execution.
4. Update your claim boundary to match the pathway, and mark the change as a new Contract version.

## A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

## The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** A change of pathway can change your permission status; recheck rather than assume.
- **Evidence, provenance, and reproducibility.** Name the prior work that used this pathway on a comparable question.
- **AI activity, verification, and human decisions.** Ask an assistant to argue for a different pathway, then answer the argument yourself.
- **Uncertainty, claim boundary, and revision history.** Different pathways carry different sources of uncertainty; name yours here.

## Where this milestone sits

You arrived carrying Milestone 4: Your research contract, v0. Milestone 6 puts data and measurement under governance so the declared pathway has something trustworthy to run on. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

## What sends you back

Realized data that will not support the declared pathway sends you back here before analysis, not after.

## The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0, 1, or 2. 10 points total.**

| # | Criterion                                                             | 0                                                  | 1                                                                              | 2                                                                       |
|---|-----------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | The milestone artifact exists in full, specific to your own project   | absent, or generic text that would fit any project | present but incomplete, or specific in some parts and generic in others        | complete and unmistakably about your project                            |
| 2 | The milestone is written as a dated, numbered version with its reason | no version, or a version with no reason given      | versioned, but the reason is decorative rather than an account of what changed | versioned with a reason a reader could use to reconstruct your thinking |

| # | Criterion                                                                                                                                                                                                         | 0                                                                   | 1                                                         | 2                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------|
| 3 | Every judgment in the artifact is defended by you, not asserted by a tool                                                                                                                                         | conclusions appear with no reasoning, or reproduce a tool's wording | reasoning present but thin where the decision was hardest | each judgment carries a reason you could defend under questioning |
| 4 | The four rails are carried in the form this studio requires                                                                                                                                                       | one or more rails absent where this studio required them            | all rails present but one is nominal                      | every rail addressed in the form this studio requires             |
| 5 | The milestone's own product is present and defensible: Research Contract v1: objective by target and reach by data strategy by warrant, with the pathway declared and its limits written before any result exists | the milestone's own product is missing                              | present but would not survive a careful question          | present and defensible as stated                                  |

## Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

## Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

### 1. Prepare the notebook

1. **Run the whole thing.** **Runtime** → **Run all**, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

### 2. Print it to a file

**File** → **Print** opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

### 3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

## One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.